

Durability of Glass Fiber-Reinforced Polymer Reinforcing Bars in Concrete Environment

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Abstract: Accelerated aging tests are being conducted on more than 20 types of glass fiber-reinforced polymer (GFRP) reinforcing bars, which are produced from different combinations of constituent materials, manufacturing parameters, sizes and shapes, and surface coatings. The specimens are being subjected to various sustained tensile loading (22 to 68% of ultimate strength) in three types of alkaline environments: NaOH, simulated pore-water solution, and embedded in concrete. Time to rupture or residual strength, as applicable, have been determined. Additionally, stress corrosion mechanisms were evaluated by various microstructural analyses. The results showed clearly that alkaline ions and moisture could penetrate or diffuse through the resin (or through cracks and voids) to the interphases and the fibers. For GFRP bars embedded in moist concrete under various sustained stress levels, three types of stress corrosion mechanisms have been identified: stress dominated, crack propagation dominated, and diffusion dominated.

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Introduction

Glass fiber-reinforced polymer (GFRP) materials have emerged as a practical alternative material for producing reinforcing bars for nonprestressed concrete structures (Benmokrane and Wang 2001). This is due to their relatively low cost-to-performance ratio and noncorrosive nature. Additionally, GFRP materials exhibit several properties, such as high tensile strength, that make them suitable for use as structural reinforcement. However, their durability in an alkaline environment is still of concern. The internal concrete environment has high alkalinity and moisture with a pH between 10.5 and 13.5 (Diamond 1981; Taylor 1987). Depending on the design mixture of the concrete and type of cement used, this alkaline environment can damage glass fibers through loss in toughness and strength and through embrittlement. Fibers are damaged due to a combination of two processes: (1) chemical

attack on the glass fibers by the alkaline cement environment; and (2) concentration and growth of hydration products between individual filaments (Murphy et al. 1999). The embrittlement of fibers is due to the nucleation of calcium hydroxide on the fiber surface. In addition, calcium, sodium, and potassium ions found in concrete pore solution are highly aggressive toward glass fibers. Ros-tasy (1997) reported that in a pH 13 environment, a significant loss of short-term tensile strength of GFRP occurs within 1,000 h of storage in 1N NaOH and cementitious solution at 23°C.

Although resin matrices provide a certain level of protection to fibers from this degradation, migration of high-pH solutions and alkali salts through resin (or through a void, crack, or fiber-matrix interface) to the fiber surface is possible (Katsuki and Uomoto 1995). Therefore, significant research is being conducted to develop new types of glass fibers such as Advantex and alkali-resistant (AR) glass fibers, apply special surface coatings, select suitable chemistry sizing, and improve the manufacturing process as well as the durability of GFRP reinforcing bars in an alkaline environment.

Besides the degradation mechanism of GFRP bars in an alkaline environment, studies also have been conducted to develop accelerated aging procedures and predictive models for long-term strength estimates. Several models are being developed for accelerated aging tests of GFRP composites, such as the glass reaction model developed at Iowa State University (Porter et al. 1995) and diffusion models developed at West Virginia University and the University of Southern Florida (Faza et al. 1994). Research on the effects of temperature on the durability of fiber-reinforced polymer (FRP) bars in a concrete alkaline environment indicates that an accelerated factor for each temperature difference can be defined by using Arrhenius laws. These factors differ for each product, depending on type of fiber, type of resin, and bar size. In addition, these factors are affected by environmental conditions such as the surrounding solution media, temperature, pH, moisture, and exposure conditions (Benmokrane and Rahman 1998; Porter and Barnes 1998). Gerritse (1998) indicated that at least three elevated temperatures were necessary to perform an accu-

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rate Arrhenius kind of prediction, and the measured data should be in continuous time intervals.

Despite a significant amount of research being conducted in this area, it is still difficult to assess the durability performance of GFRP reinforcing bars in alkaline environments. More specific aspects include (1) the degradation mechanism of GFRP bars with different combinations of constituent materials (fiber, resin, sizing chemistry, additives, and fillers) in an alkaline environment and subject to sustained load; (2) the synergistic effects of stress, moisture, and alkalinity; and (3) a reliable test methodology, suitable quality control, and quality assurance protocol.

The study reported in this paper is to evaluate degradation mechanisms of GFRP bars in three types of alkaline environments (NaOH solution, simulated pore-water solution, and embedded in moist concrete) and subject to sustained axial tensile load (22 to 68% of ultimate tensile strength) in order to identify the critical durability influencing factors, that is, the effect of fiber type, resin type, manufacturing parameters, surface coatings, and suitable combinations of these factors. This will assist industrial companies to improve the characteristics of their FRP products (types of resins and fibers, manufacturing processes, surface coating, and so on), enhance their long-term durability properties, and help engineers and users to utilize this innovative technology with enhanced confidence.

Experimental Program

More than 400 accelerated aging tests were conducted on GFRP reinforcing bars. GFRP bar specimens were subjected to various sustained loads of 25 to 68% of the ultimate tensile strength in various alkaline environments for a specific period of time. The time to rupture of specimens was recorded, and if no rupture occurred during the tests, the residual tensile strength of the aged rebar was evaluated. More specifically, 60 tests have been conducted on GFRP bars conditioned in NaOH solution under a stress level of 22 to 38% of ultimate strength; 180 tests have been conducted on GFRP bars conditioned in NaOH solution and simulated pore-water solution and embedded in moist concrete under stress level of 30%; and 110 tests have been conducted on GFRP bars embedded in moist concrete under a stress level of 25 to 68%.

Furthermore, microstructural analyses were conducted to identify the most significant stress corrosion mechanisms, such as matrix cracking, matrix degradation, matrix diffusion, fiber damage, and fiber-matrix interface damage. These results yield valuable information on the means to improve rebar durability.

Materials

Unidirectional GFRP rebars (type HB, C, and IS) produced by three FRP manufacturers were used in this research program. They were grouped into more than 20 types of specimens with different combinations of matrix types (such as vinyl ester and polyester); glass fiber types [such as normal E-glass, E-glass 366, Advantex 366, and Advantex 712 alkaline resistant glass fiber (AR)], of which Advantex glass fiber was provided by Owens Corning (Williams et al. 1999); manufacturing parameters; sizes and shapes; and sizing chemistry. Additionally, the HB rebar was an external, helically wrapped bar [Fig. 1(a)]; the C rebar surface had lugs or protrusions (ribbed bar) [Fig. 1(b)]; and the IS rebar surface had a sand coating (sand-coated bar) [Fig. 1(c)]. The tensile strengths of these rebars varied from 530 to 960 MPa, and the

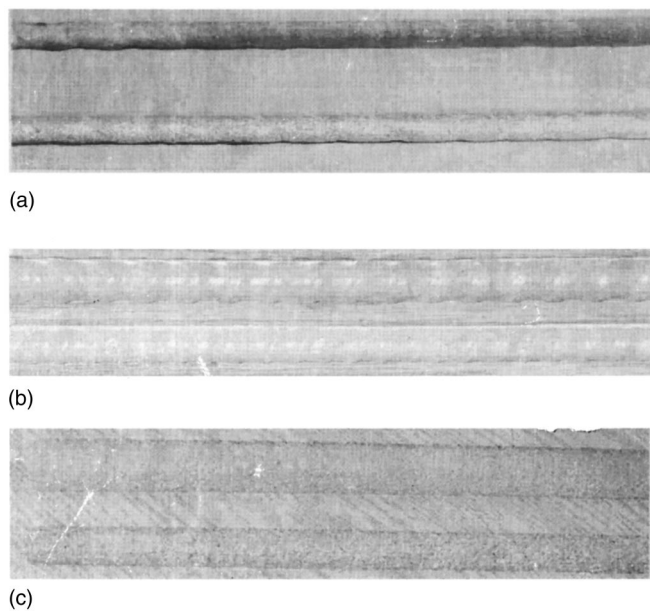


Fig. 1. Some GFRP bars used in this project: (a) typical helically wrapped rebar (type HB); (b) typical ribbed rebar (type C); (c) typical sand-coated rebar (type IS)

modulus of elasticity varied from 40 to 50 GPa. The properties of tested rebars are given in Table 1.

Development of Sustained Loading Systems

No standard method exists for evaluating the durability of FRP reinforcing rods used for concrete structures. Therefore, several test methods involving gripping and loading techniques for subjecting GFRP rebar specimens to a constant load as well as in alkaline environments have been investigated. To overcome the disadvantages of the presently existing system, new loading systems were developed, namely, creep frame loading and compressed spring loading, shown in Figs. 2 and 3.

Creep Frame Loading

Creep frame loading was developed to eliminate the loss of load, as shown in Fig. 2. In principle, a tensile load was kept constant by a gravimetric force of the counterweight. This setup was used for the tests in NaOH solution, a simulated pore-water solution at pH=12.8, and embedded in moist concrete. This setup had load control in addition to no loss in applied load. The main disadvantage of this setup is that it requires a large space.

Compressed Spring Loading

Compressed spring loading was based on a one-spring jig (Fig. 3). Due to the relatively low stiffness of the spring, only a small loss of load, with less than 0.5%, was observed, even at a high stress level of 67% over 15 days. This setup was found to be the best overall for this study as it has a very small loss in applied load and is compact. The series of specimens embedded in concrete was tested with this setup at different stress levels from 20 to 65% of the ultimate tensile strength.

Table 1. Properties of Some Studied Glass Fiber-Reinforced Polymer Bars

Rebar types	Fiber type	Matrix type	Fiber content (weight ratio)	Diameter (mm)	Manufacture processing	Tensile modulus (GPa)	Tensile strength (MPa)
HB1	E-glass 366	P	73–78%	9.3	MP1	—	720±57
HB2	E-glass 366	5P 2V		12.4	MP1	46.7±1.1	702±23
HB3	E-glass 366	1P 1V		12.4	MP1	51.2±2.3	634±9
HB4	E-glass 366	V		12.4	MP1	51.8±1.3	612±16
HB5	AR glass	P		9.3	MP1	31.6±0	650.1±3.7
HB6	AR glass	1P 1V		9.3	MP1	34.5±0	617.1±6.9
HB7	AR glass	V		9.3	MP1	41.7±6.8	531.8±11.9
HB8	E-glass	V		9.3	MP1	35.4±0.6	958.3±5
C1	E-glass	V-PET	63–73%	12.0	P1	34.1±1.3	658±36
C2	E-glass	V-PET		12.0	P1	37.4±0.7	693±11
C3	E-glass	V		12.0	P2	35.0±0.9	659±21
C4	E-glass	V		12.0	P3	35.2±1.2	669±42
C5	E-glass	V-PET		12.0	P1	42.0±0.9	577±50
C6	E-glass	V		12.0	P2	35.3±0.8	624±19
C7	E/A1	V		12.0	P2	38.3±1.4	660±7
IS1	E-glass	P	71–76%	9.5	IP1	41	651
IS2	E-glass	V1		9.5	IP1	39	689
IS3	E-glass	V1		9.5	IP2	38	578
IS4	E-glass	V1		9.5	IP1	35.5±1.0	527±18
IS5	E-glass	V2		12.4	IP1	—	—
IS6	E-glass	V2		9.5	IP2	34.9±2.7	580±21
IS7	Carbon	V2		9.5	IP1	31.8±0.1	534±30

Note: P=polyester resin; V, V1, V2=vinyl ester resin, vinyl ester 1, vinyl ester 2; Ar=alkaline resistant glass fiber; MP1, P1 P2, P3 IP1, IP2 = manufacturing methods.

Simulated Alkaline Environments

The next important issue for the tests was to simulate the alkaline environment of concrete. At first, the NaOH solution (pH 12 to 13.5) was used, but even NaOH, one of the strongest alkalis, is not the real alkaline composition of the pore-water solution; it was used to accelerate a degradation of the specimens in order to choose the most alkaline-resistant bars for the later tests.

A simulated pore water solution was considered in the second series. The high pH of pore-water solution was created through a series of chemical reactions in concrete during the hydration processes that were produced mainly by calcium hydroxide $\text{Ca}(\text{OH})_2$ and small amounts of sodium and potassium hydroxides (Nilson and Winter 1991). The composition of pore-water solution depends on cement type and can vary widely. The simulated pore-water solution for this study was prepared from a mixture of 118.5 g of $\text{Ca}(\text{OH})_2$, 0.9 g of KOH in 1 L of deionized water,

which is the most common solution for FRP bar durability tests (Benmokrane and Rahman 1998). The final pH was found to be about 12.8 at room temperature.

Degradation of FRP reinforcements in concrete is a complicated process involving various chemical reactions. For the short term, degradation of the GFRP bars can be due mainly to reactions with alkali, but for the long term, reactions with other components of the pore-water solution should also be taken into account. An example of this reaction is oxidation accelerated by



Fig. 2. Loading system 2

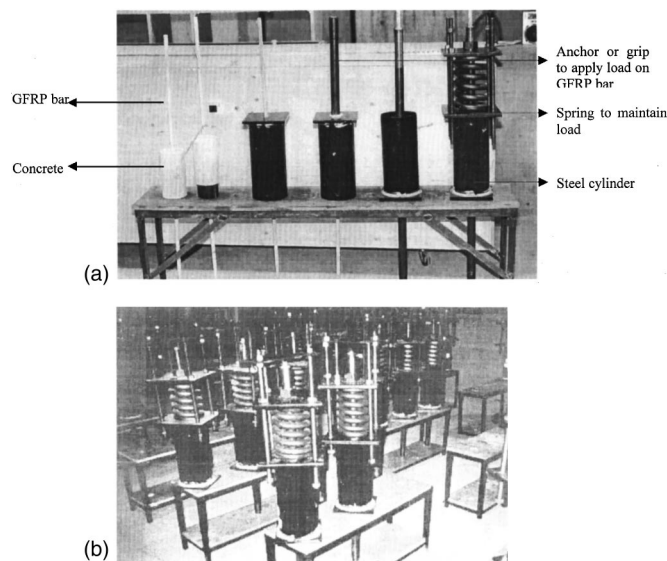


Fig. 3. Loading system 3: (a) setup steps for loading system 3; (b) close view

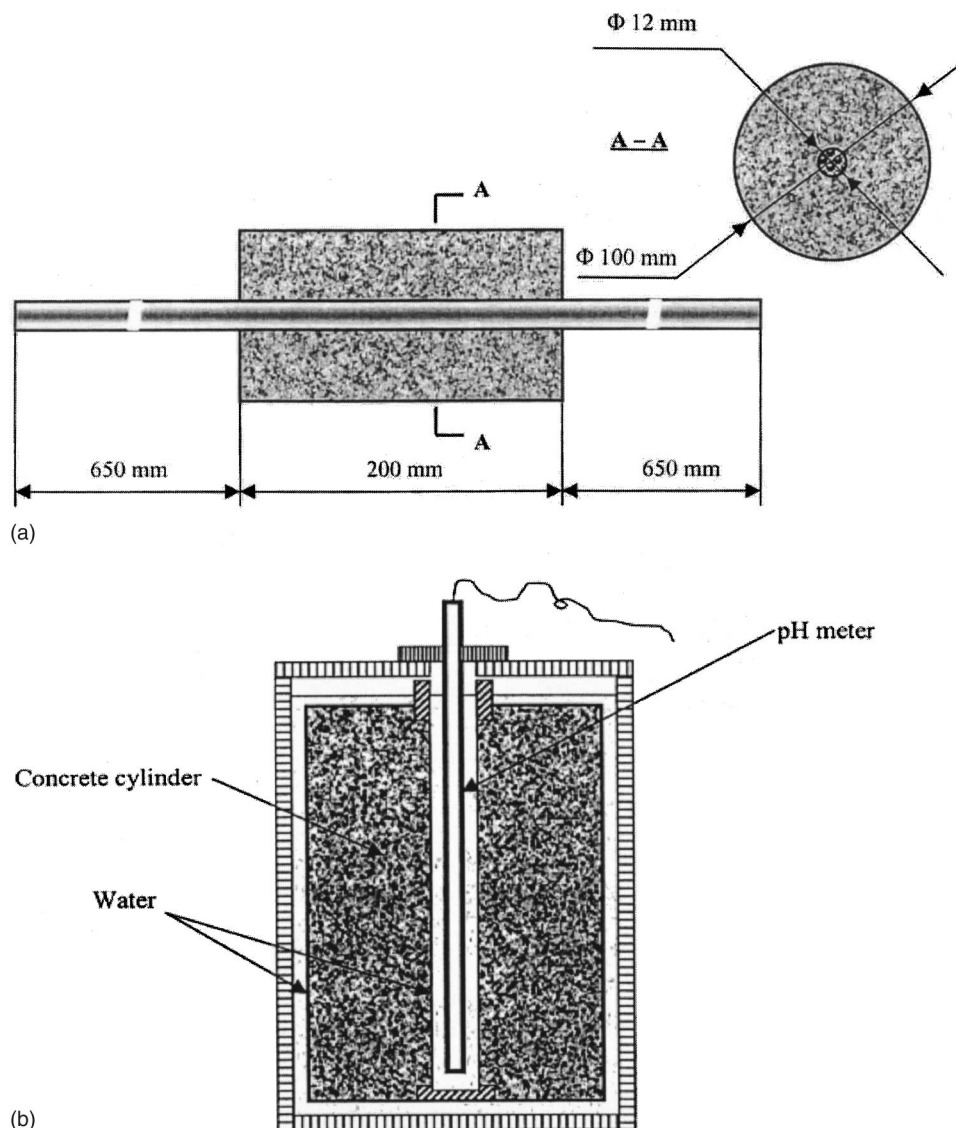


Fig. 4. Test setup and specimen for determination of pH in concrete: (a) specimen; (b) test setup

heavy metal ions of pore-water solution. For a realistic evaluation of the durability of GFRP bars in concrete, it was necessary to do the tests with specimens embedded in concrete. In the last series, the bar was embedded in concrete [Fig. 4(a)], made with a cement of CSA Type 10 (ASTM Type 1). The concrete-embedded specimens were kept in the water in which they had been cured at room temperature for 2 months before the test [Fig. 4(b)]. This water was also used to immerse the specimens during the test; its pH was found to be around 10.5.

To control the pH at the concrete cylinder, an additional experiment had been carried out. Five similar concrete cylinders were prepared with a hole of 12 mm in diameter at the center. The hole was closed at one end and a hollow cylinder was fixed into the hole at the other end. After 24 days curing in humid air of 100% relative humidity at room temperature, the cylinder was immersed in a water container. After 30 days, the pH at the hole was determined to be 13.1.

Material Characterization

Prior to initiating the test program, the physical properties of GFRP bars, which include fiber content, thermal stability of resin,

fiber distribution, and cross-sectional area, were characterized. Fiber distribution and perfection in manufacturing of the rebar were studied by scanning electron microscopy (SEM). Thermogravimetric analysis (TGA) was used to evaluate resin/fiber content and estimate the thermal stability of the resin matrix via the temperature at a weight loss of 5% (T_5) and of 10% (T_{10}), the weight loss at 500 and 800°C (W_{500} and W_{800} , respectively) and the decomposition peak temperature (T_d) (at which the decomposition rate is the maximum, determined by the maximum peak of the differential thermal gravimetric curve). As the rebar shape is not circular and uniform along the axis, the rebar cross-sectional area is determined by the immersion method. Additionally, absorption tests have been conducted on GFRP bars immersed in distilled water and NaOH solution at room temperature and 60°C. The absorption tests were carried out according to the ASTM D543-87 standard test method.

Tensile Properties

FRP bars cannot be tested in tension using the same techniques as steel bars. If a FRP bar is loaded using traditional wedge-shaped frictional grips, the combination of high compressive stresses and

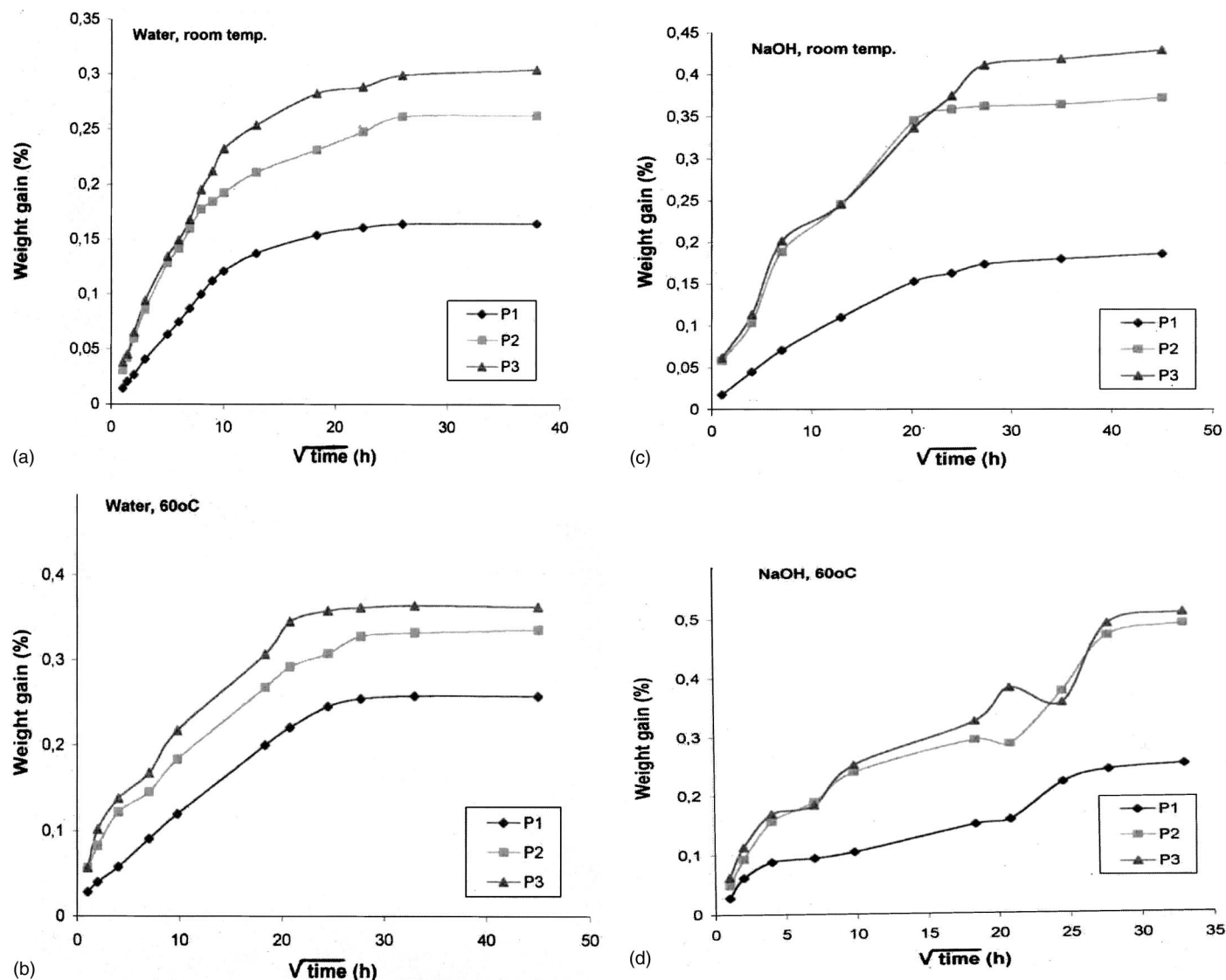


Fig. 5. Typical absorption curves for samples immersed in different solutions and temperatures: (a) water at room temperature; (b) water at 60°C; (c) NaOH (pH 13.1) at room temperature; (d) NaOH (pH 13.1) at 60°C

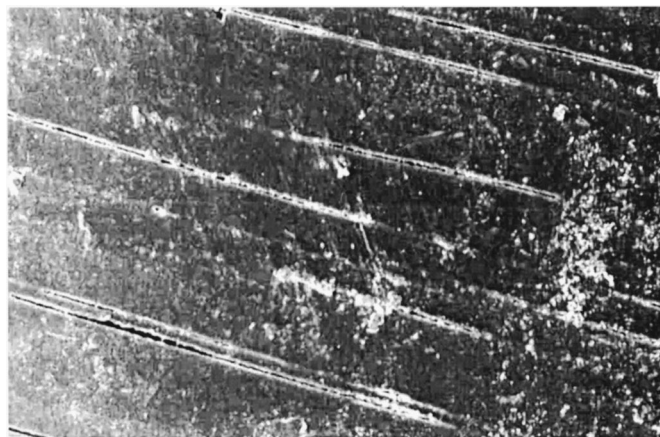


Fig. 6. Surface damage of P1 specimen in NaOH solution at 60°C for 30 days

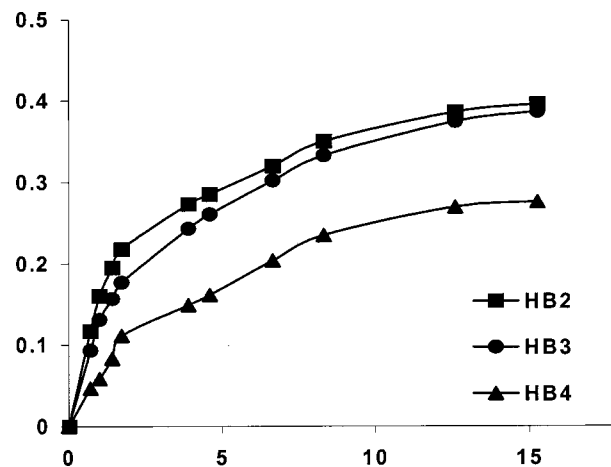
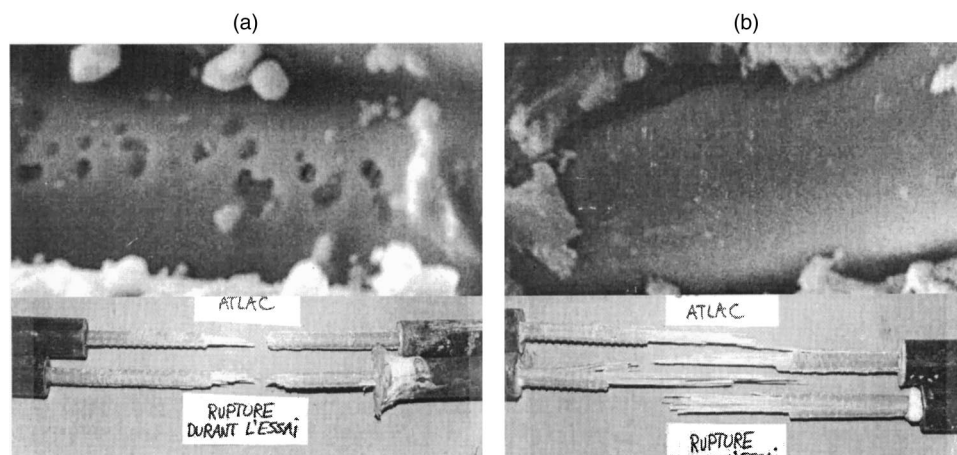


Fig. 7. Typical water absorption test on glass fiber-reinforced polymer bars

Table 2. Typical Stress Corrosion Test Results for Glass Fiber-Reinforced Polymers Bars after Immersed in Simulated Pore Water Solution under 22°C

Rebar type	Number of specimens	Stress levels (% of tensile strength)	Number of days under tension	Average retention in tensile strength (%)
HB2	3	30	90	84.8±5.1
HB3	3	30	90	85.6±3.5
HB4	3	30	90	96.0±1.1

**Fig. 8.** Typical scanning electron microscope photos for fiber surface of aged specimen in pore water: (a) C6 specimen; (b) C7 specimen**Table 3.** Typical Residual Strengths of Glass Fiber-Reinforced Polymers Bars Aged in Different Alkaline Environments under 22°C

GFRP type	Number of specimens	Alkaline environments	Stress level (%)	Number of days	Diameter (mm)	Residual strength (MPa)	Changes in strength (%)
HB5	5	In simulated pore water	30	140	9.3	618.3±35.3	-5
HB6	5		30	140	9.3	624.5±0	+1
HB7	5		30	140	9.3	537±22.5	+1
HB8	5		30	140	9.3	797.9±15.5	-17
HB5	5	Embedded in concrete	30	140	9.3	659.9±23.6	+2
HB6	5		30	140	9.3	621±5.4	-1
HB7	5		30	140	9.3	525.6±9	-1
HB8	5		30	140	9.3	804.3±22.1	-16

mechanical damage caused by the serrations on the wedge surface will lead to premature failure of the grip zone. The grip developed in this study is a steel cylinder filled with epoxy resin. The metal grip was also used for carrying sustained load during the test over a long period of time in a highly corrosive environment.

Results and Discussions

Diffusivity Behavior

Fig. 5 gives the absorption curves of three different types of bars—C5, C4, and C6—produced by different manufacturing processes—P1(C5), P2(C6), and P3(C4)—in various environments and temperatures. It is clearly seen that the absorption behavior of all specimens is similar. At the beginning, the absorption rate was high; the rate then decreased with time as the water concentration gradient between the surface and the inner part of the specimen decreased.

At the early stage of absorption, the P1 curves in most cases are fairly smooth while the P2 and P3 are not. The non-Fickian characteristics of P2 and P3 could be explained by the presence of holes as well as voids and microcracks, either preexisting or cracked by moisture/alkalis during the absorption. The nonlinear absorption for P1 found in the test in NaOH at 60°C could be explained by the degradation at its surface, as indicated in Fig. 6.

It can also be observed from Figs. 5(a–d) that the moisture uptake in water is slower than in NaOH, and the absorption rate increases with temperature. The faster diffusion in NaOH is clearly due to the reaction between NaOH and ester linkages of resin. As movement of the materials becomes faster at high temperature, it is easy to understand that the diffusion of water and alkaline molecules into bars increases with temperature.

Effect of Resin Matrix

The resin matrix protects the fibers from mechanical and environmental attacks, ensures that the fibers remain aligned, and allows

Table 4. Typical Results for Effects of Manufacturing Parameters in Stress-Rupture Tests under 22°C

Rebar type	Manufacture parameter	Number of specimens	Stress level (% of ultimate tensile strength)	Number of days at rupture	Note
C2-1	P1	5	64	90	
C2-2	P1	5	65	83	
C2-3	P1	5	65	176	Out of weakest part
C2-4	P1	5	65	198	Out of weakest part
C3-1	P2	5	65	193	
C3-2	P2	5	65	248	

loads to be transferred to the individual fibers. The most interesting properties of the resin matrix for use in FRP reinforcing bars are their chemical stability, diffusivity, and toughness. Typical stress corrosion and absorption tests on two resin types (polyester and vinyl ester) or combinations (for bars HB2, HB3, and HB4), obtained from the same manufacturer and produced by the identical manufacturing technique, are reported as follows.

The absorption behavior of three bars can be seen in Fig. 7. Diffusion of water was faster in specimen HB2 than HB3 and was slowest in HB4. Diffusibility of FRP depends on the resin and fiber content. Higher fiber content should lead to low diffusibility since the fiber is considered negligibly permeable. As the three rebars have almost the same fiber content, type of resin must be the factor responsible for the observed difference in diffusion. Vinyl ester resin contains fewer polar groups than polyester (Springer 1984), and hence vinyl ester and its composites should show low diffusibility and absorbability. Besides, polyester has reactive double bonds distributed throughout the chains, while vinyl ester has such bonds at the ends of the chains. Thus the cross-link density can be better controlled in vinyl ester (Riffle et al. 1998). As a result of good cross-links, vinyl ester resin should be more stable and better able to resist water penetration, which is the reason for the lower water absorption of the rebar HB4, which is made of vinyl ester resin.

The effect of the resin type on the stress-corrosion characteristics of the rebars was also investigated. From Table 2 it can be seen that the tensile strength of all rebars slightly decreased after aging in simulated pore-water solution under a stress level of 30% for 90 days at 22°C. The results clearly show that the durability of GFRP bars was influenced by the matrix used. The rebar HB4 made of 100% vinyl ester resin exhibited the highest resistance to stress corrosion in alkali. It is well known that vinyl ester resin has better chemical resistance, even in an alkaline environment. Furthermore, vinyl ester resin is tougher than polyester because of a longer distance between cross-links. Thus, if flaws/microcracks

preexist in a specimen, their growth under stress would be at a lower rate in vinyl ester than in polyester.

Effect of Fiber Types

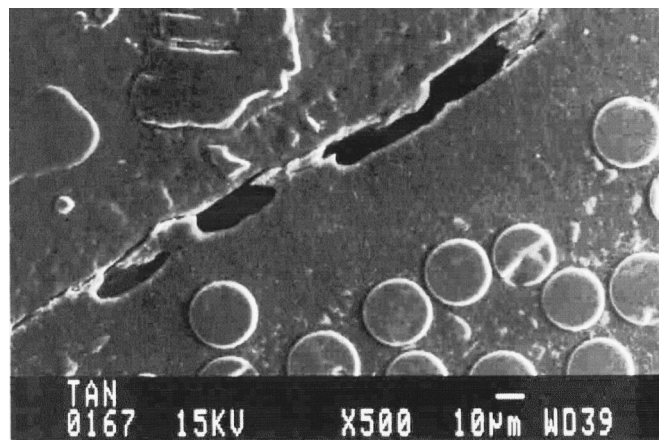
Fibers are the main load-carrying element for unidirectional FRP reinforcing bars. Obviously, degradation in fibers due to aggressive environments could lead to significant reduction in the strength and stiffness of FRP reinforcing bars. Typical test results on alkaline resistance of E-glass (aluminoborosilicate), Advantex, and AR glass fibers were reported.

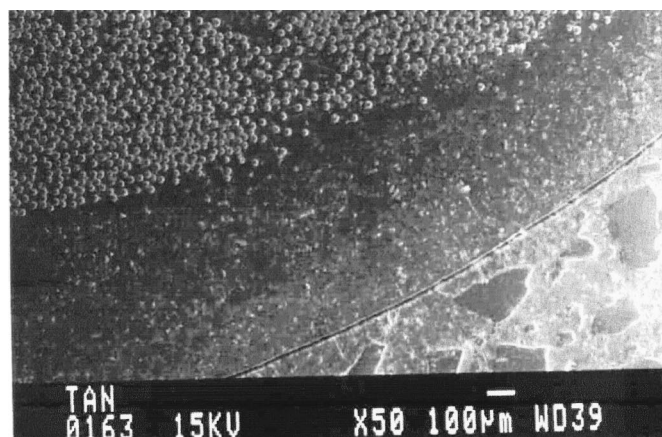
GFRP bar specimens were immersed in simulated pore-water solution and kept at the same constant stress level of 30% for 140 days. A SEM analysis was conducted on two specimens, C6 and C7, from the outer surface of the bars, which were produced with the same resin matrix but different glass fiber types. More fiber damages were observed in the C6 than in the C7 specimen (Fig. 8). Therefore, C6 showed a flat fracture while C7 appeared to have a more broomlike fracture. The difference in failure morphologies between the two is due to the glass fiber used: C7 was based on Advantex, and C6 on normal E-glass fiber. This indicates that Advantex is more alkaline resistant than normal E-glass.

Table 3 shows typical stress corrosion test results on AR/vinyl ester and E-glass/vinyl ester bars after being immersed in simulated pore-water solution and embedded in concrete under a 30% stress level for 140 days. AR/vinyl ester bar showed almost no reduction in tensile strength in two simulated accelerated aged conditions, while E-glass/vinyl ester showed 16 and 17% reductions in tensile strength after being aged in two accelerated aging conditions. Obviously, in the current study, GFRP bars reinforced

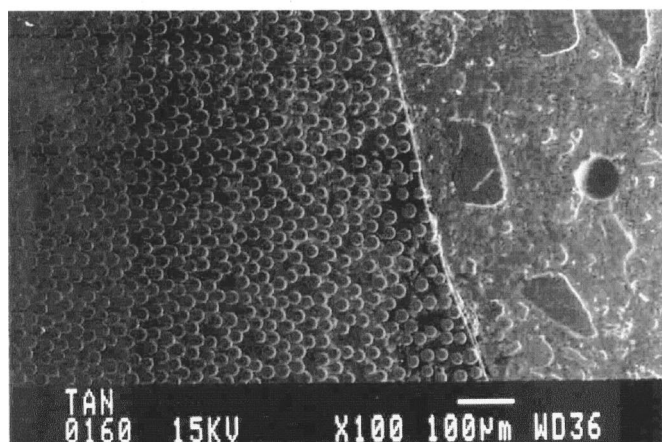
Table 5. Typical Results of Stress Corrosion Test on Glass Fiber-Reinforced Polymer (GFRP) Bars Embedded in Moist Concrete under 22°C

GFRP bar type	Number of specimens	Stress level (%)	Duration (days)	Number of days at rupture
IS0	18	21–44	1,295–1,445	Ongoing
IS4	14	25–39	950–1,013	Ongoing
IS4	10	48–65	—	1–27
HB2	14	24–47	1,086	Ongoing
HB2	14	46–66	—	1–135
HB3	14	19–37	1,232	Ongoing
HB3	13	54–68	—	3–34

**Fig. 9.** Scanning electron microscope images of interface between fiber-reinforced polymer rebar and concrete



(a)



(b)

Fig. 10. Scanning electron microscope images of interface between glass fiber-reinforced polymer and concrete: (a) at resin-rich zone; (b) at fiber-rich zone

with AR glass fibers show better alkaline resistance than the GFRP bars reinforced with E-glass fibers.

Effect of Manufacture Parameters

The effect of manufacturing parameters on durability was illustrated by a study of the C rebars. Rebar C2 was manufactured in a condition described as P1, while rebar C3 was manufactured in

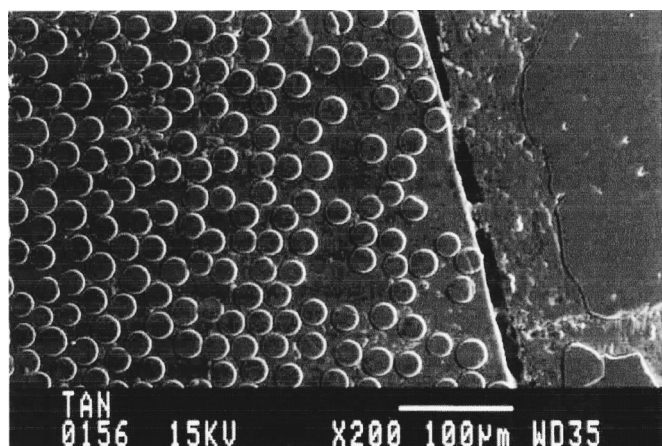


Fig. 11. Failure mode of specimen aged in moist concrete

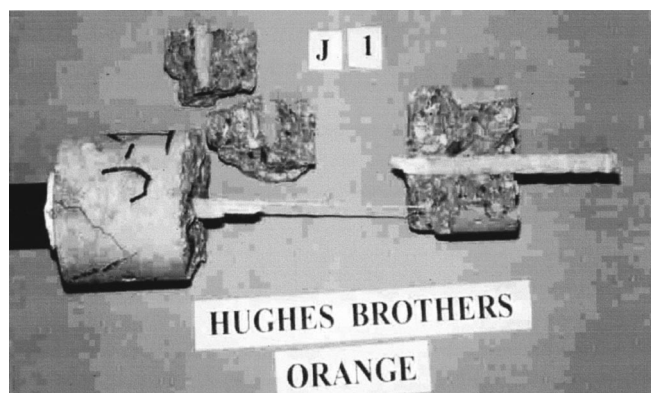


Fig. 12. Typical stress corrosion test results for glass fiber-reinforced polymer (GFRP) bars embedded in concrete: (a) typical rupture life and exposure time of GFRP bars; (b) typical rupture life of GFRP bars

P2. Two rebars were subjected to a NaOH solution (pH=13.1) under a stress level of 30% for a duration of 60 days. No significant difference in their residual tensile strength could be observed. However, when bars were subjected to the high stress level of 65% in air, the results indicated a difference (Table 4). The P1 impaired a weakness in the rebar C2 at the location of the mould joints. Rupture of bar specimens C2-1 and C2-2 began at this position and then developed further. This was a possible reason for a faster rupture of these specimens, as seen in Table 4. For

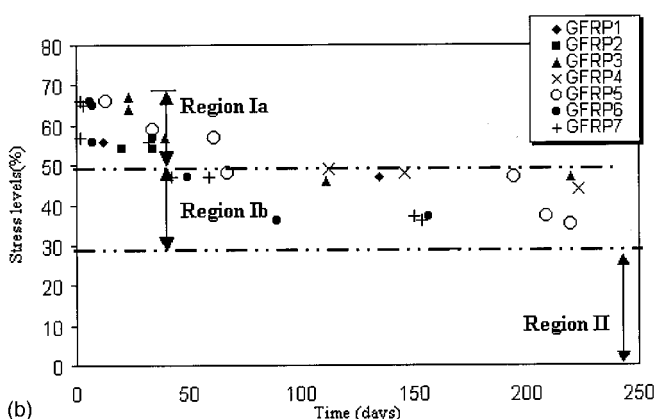
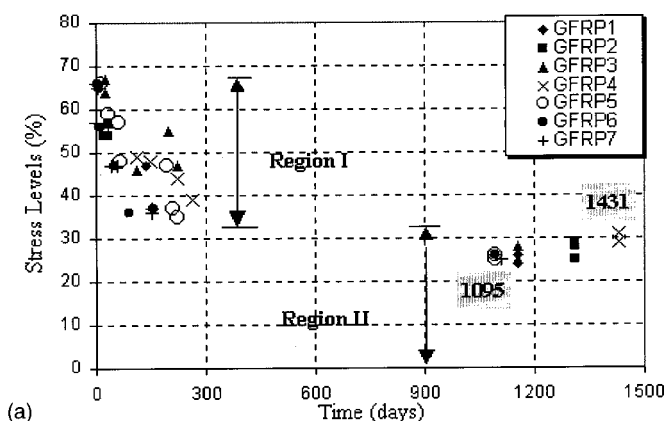


Fig. 13. Typical alkaline penetrations at wrapping areas: (a) thermogravimetric analysis specimen; (b) scanning electron microscope images

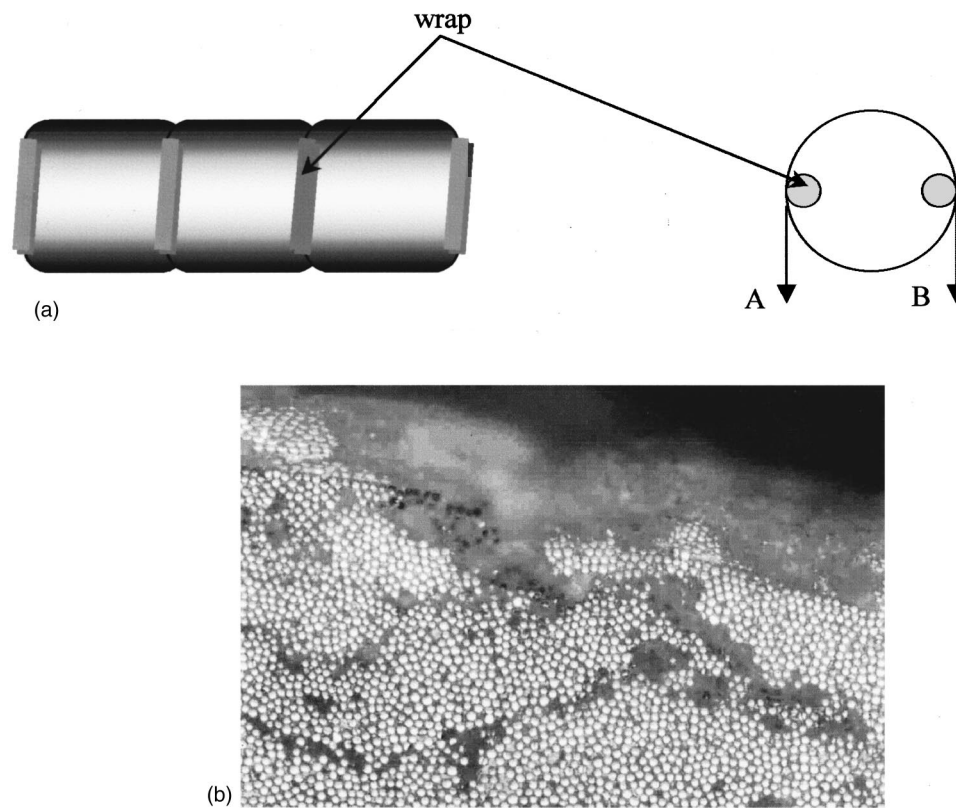


Fig. 14. Typical alkaline penetrations at wrapping areas: (a) thermogravimetric analysis specimen; (b) scanning electron microscope images

specimens C2-3 and C2-4 that did not have this part within their lengths, the time to rupture is significantly longer, as seen from Table 4. It appears that P2 could rectify the problem of P1 as rupture of specimens C3-1 and C3-2 took longer.

Stress Rupture Tests on Glass Fiber-Reinforced Polymers Bars Embedded in Moist Concrete

GFRP reinforcing bars embedded in concrete were kept in concrete-curing water at room temperature for 2 months before being subjected to different stress levels from 25 to 68% using loading system 3. The intention of this stress rupture test is to investigate the stress corrosion mechanism of GFRP bars in moist concrete and establish the safe operating stress level for a given GFRP reinforcing bar system exposed to moist concrete. The test will be continued for 10 years, and time of rupture or residual strength are being recorded. Some preliminary results are shown in Table 5.

SEM investigations on the cross section show that there is a good bonding between concrete and bar surface. However, at high magnification, a few gaps at the composite-concrete interface can

be observed (Fig. 9). Because of a high surface tension between the concrete (hydrophilic) and composite (hydrophobic) surfaces, holes on the surface of specimens, due to imperfections in the manufacturing process, cannot be filled up by the cement or concrete (Fig. 9).

When the test specimen was subjected to a stress level of 30% of the ultimate strength for 30 days, a debonding at the concrete-bar interface was observed as a gap appeared at this interface. Since the resin-rich zone (surface deformations) has a higher Poisson ratio, transverse shrinkage due to tension in this region is higher, resulting in a larger gap at the interface [Fig. 10(a)]. At the fiber-rich zone, only a narrow gap was observed [Fig. 10(b)]. The gap becomes greater at a higher applied load of 50% due to a larger deformation. Interface damage and matrix cracking were observed (Fig. 11).

More rupture took place within the embedded part than in the free zone. In all aged and unaged specimens, “broomlike” fracture failures are similar, including longitudinal splitting and transverse cracking. However, in the aged specimens transverse cracks were joined by longitudinal ones with far less longitudinal splitting, and failures occurred over a short distance (Fig. 12).

Table 6. Typical Residual Strength of Glass Fiber-Reinforced Polymers (GFRP) Bars Aged in NaOH Solution under 22°C

GFRP bar types	Number of samples	Stress level (% of ultimate strength)	pH	Number of days under tension	Average retention in tensile strength (%)
C2	10	30	12.1	30	98
C2	10	30	13.1	30	94
C2	10	30	12.1	60	100
C2	10	30	13.1	60	78

Threshold Stress Level

Fig. 13(a) describes typical results for seven kinds of GFRP bar specimens embedded in moist concrete under different stress levels (25 to 68%). The differences in these GFRP bars are fiber and resin types, fiber resin/weight ratio, and manufacturing processes. Therefore, results presented in Fig. 13(a) could provide a general tendency of the stress corrosion mechanism for GFRP bars. Data in stress level region I (35 to 68%) describe rupture life of the specimens under different stress levels. Data in stress region II (25 to 30%) describe the up-to-date measurements on specimens that are not ruptured and are still under sustained tension load; some specimens are still under tension between 1,095 and 1,431 days without any rupture. Clearly there are threshold stress level regions at which the microcracks of the resin will not occur to the extent where capillary action will dominate the degradation rate of the specimens. Stress corrosion of the specimens under these threshold stress level regions was found to be diffusion dominated, and failure of the specimens was less stress sensitive. The fundamental reason for this threshold stress level is due to the viscoelastic behavior of polymeric resin; at lower stress levels, the viscoelastic behavior of polymeric resin allows stress relief and crack formation does not occur. The lifetime of the specimen is determined by the rate of moist and alkali diffusion and fiber corrosion. In the present study, the threshold stress regions were found to be 25 to 30% of the ultimate tensile load. Most of the specimens were durable under this stress-level region.

To clarify the stress corrosion mechanisms of GFRP bars in region I [Fig. 13(a)], the maximum time scale in Fig. 13(a) was changed from 1,500 to 250 days; clearly stress corrosion mechanisms in region I (35 to 68% stress levels) [Fig. 13(a)] can be divided into two types of stress corrosion mechanisms, which were described as regions Ia (50 to 68%) and Ib (35 to 50%) in Fig. 13(b). Obviously, failure of the specimens in region Ia is mainly stress dominated, and specimens show brittle fracture that occurred in a short time. The failure mechanisms include fiber degradations, matrix crack, and interface debonding. Failure of the specimens in region Ib was crack propagation dominated. The rupture life of the specimens depends on the stress levels, aggressive ion concentrations, manufacture quality, and toughness of the resin matrix. Other typical results of stress corrosion tests on GFRP bars embedded in moist concrete are presented in Table 5.

Diffusion of Moist and Alkali Ion

As discussed above, in the stress rupture test, moisture diffusion dominated the failure of GFRP reinforcing bars under a lower stress level. Therefore, it is more significant to understand moisture diffusion mechanisms under this stress level for design purposes.

An imperfect manufacturing process may lead to voids, holes, microcracks, or fiber-rich zones in which moist and alkali ions could easily penetrate, causing damage of the fiber-matrix interphase and corrosion of the glass fibers. These manufacturing imperfections were commonly found in some surface deformations (that is, helically wrapped, sand coated, lugs, or protrusions). For a typical wrapped bar, thermal gravimetry analysis (TGA) measurements showed that fiber content was higher in near wrap [Fig. 14(a), position A] than at the opposite part [Fig. 14(a), position B]. This higher fiber-rich zone occurred because resin was squeezed out by the compression of the wrapping. Since surface deformations will directly contact alkaline environments, the fiber-rich zone cannot be well protected by resin, alkali could easily penetrate into the bar in this zone [Figure 14(b)]. The fail-

ure of the aged specimens was found to start at some points around wrap and then developed along this wrap.

Table 6 shows the typical residual tensile strength of GFRP bar (type C) after immersion in NaOH with different concentrations and time. Results indicated that tensile strength did not change significantly when specimens were tested at a pH as low as 12.1 during a short period of 30 days. However, the tensile strength decreased when they were tested at a higher pH of 13.1 during a period of 30 days. A more significant reduction was noticed when they were tested at a high pH of 13.1 during a period of 60 days. The observed results show clearly that penetration of alkaline ions into GFRP bars depends on concentrations, composition of alkaline solution, and exposure time; as concentration increased, the rate of alkali diffusion into the bar and the rate of the alkali attack on the bar increased, which therefore led to more reduction in the mechanical properties of the aged GFRP bars.

To reduce the diffusion rates of the moist and alkali ions, different resins were applied as surface coatings. Stress corrosion tests showed that vinyl ester resin provides better protection for the bar even when a thinner layer of resin was used for surface coatings. Sand coatings were found to reduce the microcrack and macrocrack in the resin coating surface, which occurred due to chemical reaction between the coating resins and alkali ions; this significantly reduced the diffusion rates of moist and alkali ions (Benmokrane and Wang 2001). Other parameters that influence the diffusion of moist and alkali ions, such as fiber contents, exposure time, and temperature, have been studied and reported elsewhere (Benmokrane et al. 1998).

Conclusions

This study is part of a research program now in progress at the University of Sherbrooke (Quebec). The research program is not limited to laboratory studies but also includes a field study of the long-term behavior of FRP-reinforced concrete structures. Innovative sensing techniques are being used, such as fiber-optic sensors integrated in FRP reinforcements installed in a concrete structure to collect strain-response data on FRP bars. Results from the ongoing research project were presented on the effects of an alkaline environment on the durability of GFRP bars, from which the following conclusions and recommendations can be drawn:

1. The test methodology to evaluate alkaline effects on the durability of GFRP reinforcing rods should include FRP reinforcing rods embedded in or directly in contact with concrete. The concrete should be moist/wet, and the FRP bars should be under a certain amount of stress close to or over the stress level experienced in service. Loading system 2 may be used for a durability evaluation of FRP bars, and elevated temperatures can be used to accelerate aging in the laboratory. However, in order that temperatures should not exceed the glass transition temperature of the matrix and the temperature at which different mechanisms of degradation may creep in, the temperature of 60°C has been suggested.
2. Alkaline ions and moisture could penetrate or diffuse through the resin (or through cracks and voids) to the interphase and fibers and damage the FRP rebars through many mechanisms, which change depending on the type of resin matrix, sizing, stress level, pH of the alkaline environment, and manufacturing process.
3. Microstructural techniques are very useful in evaluating the short- and long-term performance and changes taking place in FRP reinforcements under stress and environmental effects (evaluation of the resin matrix, fiber, interphase, Tg,

voids and cracks, fiber volume fraction, and so on). Such analyses as those described in this paper should be considered standards for the quality control/assurance tests by FRP industrial manufacturers, engineers, and users (codes and specifications). These techniques definitely help industrial manufacturers to produce high-quality FRP reinforcing rods and thereby maintain the high standards of their products.

4. The results obtained have also shown that the type of glass fibers, manufacturing process, and additional factors determine the long-term durability of the GFRP reinforcing rods. In particular, it has been shown that Advantex and AR glass fibers have a better resistance to alkaline environments than normal E-glass. The resin and FRP reinforcing bars should be fully cured prior to their use in concrete structures. Tests have shown that GFRP bars with undercured resins present an increase in moisture and ion penetration compared to fully cured products. Voids, holes, or other imperfections on the FRP rebar surface should be eliminated. All tested GFRP bars having these imperfections have shown less durability compared to GFRP bars not presenting these imperfections.
5. Resins also play an important role in protecting the fiber and slowing the diffusion process. Vinyl ester resin (or epoxy) shows low diffusibility and high resistance to alkaline attack and is quite tough in resisting microcrack development. Therefore, it has the highest potential to protect the glass fiber in reinforcing rods from alkaline attack and should be used in manufacturing GFRP bars. Polyester resin is not recommended.

Finally, it is clear from the obtained results that GFRP reinforcing rods can be used in structural concrete considering the two following conditions: (1) the stress level in the GFRP rods should be limited under sustained factored loads at less than 25% of the guaranteed design strength (which may be taken as average ultimate tensile strength minus three times the standard deviation); and (2) it is assumed that properly specified and proven materials and manufacturing processes for FRP reinforcing rods have been used by the manufacturers.

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